Today:
- derive Kolmogorov equations for CTMC.
- examples with Kolm. equations
- stationary distributions
- generator & Dynkin martingales

Recall Kolmogorov equations: let $p_t(x,y) = P_x(X_t = y)$

then

$$\frac{d}{dt} p_t = Q p_t \qquad \text{backward}$$

$$= p_t Q \qquad \text{forward}$$

and $Q_{ij} = \begin{cases} q_{ij} & \text{if } j \neq i \\ -\lambda_i & \text{if } j = i \end{cases}$

where $\lambda_i = \sum_k q_{ik}$.

To derive each equation use Chapman-Kolmogorov:

forward:

$$p_{t+h}(x,y) = \sum_z p_t(x,z)\, p_h(z,y)$$

$$= \begin{cases} h q(z,y) + o(h^2), & z \neq y \\ 1 - h\lambda_y + o(h^2), & z = y \end{cases}$$

$$= \sum_{z \neq y} p_t(x,z)\, h q(z,y) + (1 - h\lambda_y)\, p_t(x,y) + o(h^2)$$

with $q(y,y) := -\lambda_y$,

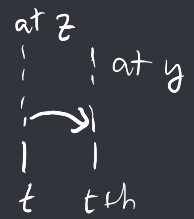
$$p_{t+h}(x,y) = h \sum_z p_t(x,z)\, q(z,y) + p_t(x,y) + O(h^2)$$

subtract $p_t(x,y)$, divide by h:

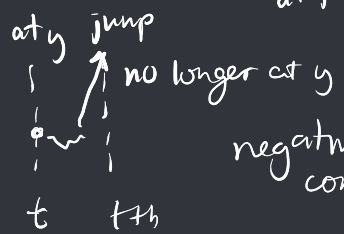
$$\frac{d}{dt} p_t(x,y) = \sum_z p_t(x,z)\, q(z,y)$$

forward equations

$$\Leftrightarrow \quad \frac{d}{dt} p_t = p_t Q$$



positive contributions $(q(z,y),\ z \neq y)$



negative contribution

(note some negative terms are needed because we must have

$$\sum_y p_t(x,y) = 1 \quad \forall x, t.)$$

ex: Poisson process w/ intensity λ.

Recall $q(n,n+1) = \lambda$ $\quad q(n,m) = 0 \quad n \notin \{m, m-1\}$

$q(n,n) = -\lambda$ $\quad$ Also $p_t(i,j) = 0 \quad \forall i > j$

(can't decrease).

If $i < j$,

$$\frac{d}{dt} p_t(i,j) = \lambda p_t(i,j-1) - \lambda p_t(i,j)$$

Let $u_j(t) = p_t(i,j)$. Then $\frac{d}{dt} u_j(t) = \lambda u_{j-1}(t) - \lambda u_j(t)$.

If $i=j$, $\frac{d}{dt} P_t(j,j) = -\lambda P_t(j,j)$

So with i fixed, $\frac{d}{dt} u_i(t) = -\lambda u_i(t)$.

take $\lambda=1$ for simplicity.

$$u_j' = u_{j-1} - u_j$$

$$u_j' + u_j = u_{j-1}$$

integrating factor

$$\frac{d}{dt}(\underbrace{e^t u_j}_{v_j}) = e^t u_{j-1}$$

$$v_j' = v_{j-1}$$

Base case $u_i' = -u_i$; $u_i(t) = e^{-t}$

$u_i(0) = p_0(i,i) = 1$ $\therefore$ $v_i(t) = 1$

$v_{i+1}' = v_i = 1$

$v_{i+1}(0) = p_0(i, i+1) = 0$

$\therefore$ $v_{i+1}(t) = \int_0^t 1 \, ds = t$

$\vdots$

$v_{i+k}(t) = t^k / k!$

$P(\text{Poisson}(t) = k)$

$= p_t(i, i+k) = u_{i+k}(t) = e^{-t} v_{i+k}(t) = e^{-t} t^k / k!$ ✓

Backward equation

$$\approx \begin{cases} hq(x,z) & z \neq x \\ (1-\lambda_x h) & z = x \end{cases}$$

$$p_{t+h}(x,y) = \sum_z p_h(x,z)\, p_t(z,y)$$

$$= h \sum_{z \neq x} q(x,z)\, p_t(z,y) + p_t(x,y)(1-\lambda_x h) + O(h^2)$$

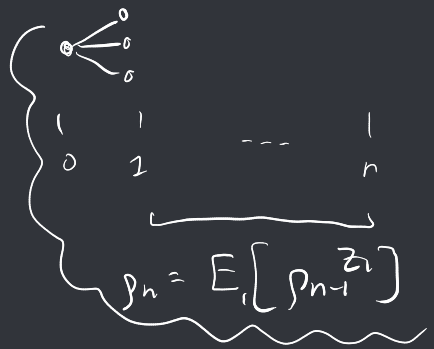
$$= h \sum_z q(x,z)\, p_t(z,y) + p_t(x,y) + O(h^2)$$

same:

$$\frac{d}{dt} p_t(x,y) = \sum_z q(x,z)\, p_t(z,y) \iff \frac{d}{dt} p_t = Q\, p_t$$

backward.

Backward equation derivation resembles one-step analysis for extinction probab. in branching process:



Can do similarly, using bkwd. equation to obtain ODE for extinction probab. in cont.-time branching process.

$$\rho(t) = p_t(1,0).$$

extinction probab.

<u>model</u>: each individual replaced @ rate 1 with k indiv. w.p. p_k.

(p_k) is specified

this gives population size Z satisfying

for each $k \in \mathbb{N}$, $k \neq 1$, $\quad Z \to Z + k - 1$ at rate $p_k Z$

i.e., $\quad q(z, z+k-1) = p_k z, \quad z, k \in \mathbb{N}$

Next compute, using blind equation,

$$\frac{d}{dt} p_t(1,0) = \sum_k q(1,k) \underbrace{p_t(k,0)}_{= p_t(1,0)^k}$$

: k independent lineages ...

$$= \sum_{k \neq 1} q(1,k) p_t(1,0)^k - \underbrace{q(1,1)}_{= \sum_{k \neq 1} q(1,k) = \cdots} p_t(1,0)$$

$$= \sum_{k \neq 1} q(1,k)\, p_t(1,0)^k - p_t(1,0) \underbrace{\sum_k p_k}_{\text{assume } = 1}$$

special case: $p_0 = p_2 = 1/2$.

(critical br. proc: $E[Z_t] = 1 \ \forall t$).

$$\frac{d}{dt} p(t) = \frac{1}{2} + \frac{1}{2} p(t)^2 - p(t)$$

$$= \frac{1}{2} \left(1 - p(t) \right)^2$$

Let $s(t) = 1 - p(t)$ survival probab.

$$\frac{d}{dt} s(t) = -\tfrac{1}{2} s(t)^2$$

Solution? IVP: $s(0) = 1 - p(0) = 1$

Solve:

$$\frac{ds}{s^2} = -\frac{dt}{2}.$$

$$-\frac{1}{s(t)} + \frac{1}{s(0)} = -\tfrac{1}{2} t$$

$$s(t) = \frac{1}{t/2 + 1/s(0)}.$$

$\searrow 0$ as $t \nearrow \infty$ but kind of slowly.

$$= \frac{1}{1 + t/2}.$$

this has $\sum k p_k = 1$. If $m := \sum k p_k < 1$

expect

$$s(t) \le C e^{-\alpha t} \quad \text{some } \alpha > 0$$

& if $m > 1$ expect

$$\lim_{t \to \infty} s(t) := s_* > 0.$$

How does m relate to mean of Z_t?

If $Z_0 = 1$ then

$$E[Z_t] = \sum_z z p_t(1, z)$$

so

$$\frac{d}{dt} E[Z_t] = \sum_z z \frac{d}{dt} p_t(1,z)$$

don't like... instead estimate

$$E[Z_{t+h} - Z_t \mid Z_t = z] = \sum_{k \neq 1} \overbrace{q(z, z+k-1)}^{\approx \text{ probability of } z \to z+k-1} h \cdot \underbrace{(k-1)}_{\text{change in value on that event}} + O(h^2)$$

$$= \sum_{k \neq 1} z\, p_k\, h \cdot (k-1) + O(h^2)$$

$$= hz \Big(\overbrace{\sum_{k \neq 1} k p_k}^{m} - \overbrace{\sum_{k \neq 1} p_k}^{=1} \Big) + O(h^2)$$

$$\frac{d}{ds} E\left[Z_{t+s} \,\middle|\, Z_t = z \right]\Big|_{s=0} = z(m-1)$$

taking $E[\cdot]$.

$$\frac{d}{dt} E[Z_t] = (m-1) E[Z_t]$$

$$E[Z_t] = e^{(m-1)t}$$

Generator & Dynkin martingale.

Let $f : S \to \mathbb{R}$, then the "generator" of a CTMC is a linear operator L defined by

$$(Lf)(x) = \sum_{y \neq x} (f(y) - f(x))\, q(x,y)$$

Lf is the "avg. rate of change" of f in the sense that

$$E_x\left[f(X_t)\right] = E\left[\int_0^t (Lf)(X_s)\,ds\right]$$

A stronger sense in which this holds is that

$$f(X_t) - \int_0^t (Lf)(X_s)\,ds$$

is a martingale (Dynkin martingale)

In particular if $(Lf)(x) = 0 \quad \forall x$

then $f(X_t)$ is a martingale.

Distribution at time t

Suppose X_0 has distribution u_0: $P(X_0 = x) = u_0(x)$.

Let $u(t) = P(X(t) = x)$.

Then $u(t) = u(0)\, p_t$ ← matrix.

row vector

$$\frac{d}{dt} u(t) = u(0) \frac{d}{dt} P_t$$

$$= u(0) Q P_t$$

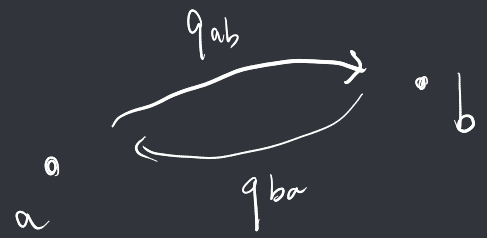
$$= u(0) P_t Q$$

A stationary distribution π is one for which $u(0) = \pi$

$\Rightarrow u(t) = \pi \quad \forall t > 0.$

From above: $\pi Q = 0$

Example:



what is stat. dist?

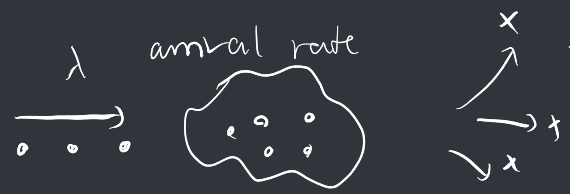
$$Q = \begin{array}{c|cc} & a & b \\ \hline a & -q_{ab} & q_{ab} \\ b & q_{ba} & -q_{ba} \end{array}$$

$$[\pi_a \quad \pi_b]\, Q = [0 \; 0] \text{ if } \quad \pi_a q_{ab} = \pi_b q_{ba}$$

$\therefore \quad \pi_a = q_{ba}/q \qquad \pi_b = q_{ab}/q \qquad q := q_{ab} + q_{ba}.$

<u>Birth-death processes:</u> CTMCs on $\mathbb{N}$: $q(i,j) = 0$ if $|i-j| \geq 2$.

example: arrival & decay



each decays indep. @ rate 1.

if X = # particles, X is CTMC with

$$q(x, x+1) = \lambda$$

$$q(x, x-1) = x$$

stat. dist.? detailed balance can be used.

stationary trans. rate $x \to x+1$

$= \pi(x) q(x, x+1)$

$$\begin{tikzcd}
x-1 \arrow[r, shift left] & x \arrow[l, shift left] \arrow[r, shift left] & x+1 \arrow[l, shift left] \arrow[r, shift left] & x+2 \arrow[l, shift left]
\end{tikzcd}$$

rates of $x \to x+1$ & $x+1 \to x$ must balance

$$\pi(x)\, q(x, x+1) = \pi(x+1)\, q(x+1, x) \qquad \forall x.$$

general formula for π of irred. b-d process

$$\pi(x) = \pi(0) \frac{q(0,1)}{q(1,0)} \frac{q(1,2)}{q(2,1)} \cdots \frac{q(x-1,x)}{q(x,x-1)}$$

for arrival / decay ex

$$\pi(x) = \pi(0) \frac{\lambda^x}{1 \cdot 2 \cdot \cdots \cdot x} = \pi(0) \frac{\lambda^x}{x!}$$

take $\pi(0) = e^{-\lambda}$ to get $\pi \overset{d}{=} \text{Poisson}(\lambda)$.

<u>exercise</u>: use short-time analysis to show $u(t) := E[X_t]$ satisfies

$$\frac{d}{dt} u(t) = \lambda - u(t).$$

☆ Hitting probability & hitting time

(X_t) CTMC. Given $A, B \subset S$: $A \cap B = \emptyset$, let

$$h(x) = P_x(X_\tau \in A) \qquad \tau = \inf\{t : X_t \in A \cup B\},$$

Then $(Lh)(x) = 0 \qquad x \notin A \cup B.$

Can use discrete time chain to see why this holds:

$$X_t = Y_{N(t)}$$

and $X_\tau \in A \iff Y_{N(\tau)} \in A$

$$N(\tau) = \inf\{n : Y_n \in A \cup B\}.$$

$\therefore$ $h(x)$ is the same for (X_t) as for (Y_n).

trans. probab. of (Y_n):

$y \neq x$: $\quad p(x,y) = q(x,y) / \lambda_x$

and from discrete theory,

$$h(x) = \sum_{y \neq x} p(x,y)\, h(y)$$

(as long as $\lambda_x > 0$).

$$\lambda_x h(x) = \sum_{y \neq x} q(x,y)\, h(y)$$

$$\sum_{y} q(x,y)\, h(y) = 0 \qquad (q(x,x) = -\lambda_x)$$